数式一覧

- k0 + k1 + k2 = 1

(1)

$$k0 + k1 + k2 = 1$$

- 初期・終端制約

(2)

$$Qr[0] + Qs[0] + Qr2[0] = Qr[48] + Qs[48] + Qr2[48]$$

- Qr 再帰制約

(3)

$$\Delta kWh = p * \Delta kW[b] * \Delta t$$

(4)

$$Qr[t+1] = Qr[t] + \eta * (1-\alpha) * \Delta kWh - \left(\frac{\alpha}{\eta * \eta}\right) * \Delta kWh$$

- Qr2 再帰制約

(5)

$$Qr2[t+1] = Qr2[t] + \eta * (1-\alpha) * \Delta kWh2 - \left(\frac{\alpha}{\eta * \eta}\right) * \Delta kWh2$$

(6)

$$Qr[t] \leq B \cdot k1[t]$$

(7)

$$Qr2[t] \leq B \cdot k2[t]$$

(8)

$$Qs[t] \leq B \cdot k0[t]$$

P_buy, P_sel 制約

(9)

$$P_{buy}[t] \leq B \cdot a[t]$$

(10)

$$P_{sel}[t] \leq B \cdot (1 - a[t])$$

Qs 再帰

(11)

$$Qs[t+1] = Qs[t] + \eta \cdot P_{buy}[t] - \frac{P_{sel}[t]}{\eta \cdot \eta}$$

x-z 関係の制約

$$x[t,0] \geq L[0] \cdot z[t,0]$$

$$x[t,0] \leq L[0]$$

$$x[t,9] \geq 0$$

$$x[t,9] \leq L[9] \cdot z[t,8]$$

$$x[t,i] \geq L[i] \cdot z[t,i]$$

$$x[t,i] \leq L[i] \cdot z[t,i-1]$$

x の合計制約

$$\Sigma x[t,i] = \frac{(P_{sel}[t] + \alpha \cdot p \cdot \Delta kW[b] \cdot \Delta t + \alpha \cdot p_2 \cdot \Delta kW2[b] \cdot \Delta t)}{B \cdot \eta \cdot \eta}$$

Qr, Qr2 に ΔkWh に基づく上下限制約を追加（t=1〜47）

$$\left(\frac{\alpha}{\eta \cdot \eta}\right) \cdot \Delta kWh \leq Qr[t-1] \leq B \cdot k_1 - f \cdot (1-\alpha) \cdot \Delta kWh$$

$$\left(\frac{\alpha}{\eta \cdot \eta}\right) \cdot \Delta kWh2 \leq Qr[t-1] \leq B \cdot k_2 - f \cdot (1-\alpha) \cdot \Delta kWh2$$

U2 の初期値制約と合計制約を時間 t ごとに定義

$$\Sigma U[i,t] = 1$$

ΣA2=1-Z

$$A[1,1,t] + A[1,2,t] + A[1,3,t] = Z'[1,0,t]$$

$$A[2,1,t] + A[2,2,t] + A[2,3,t] = Z'[2,0,t]$$

$$A[3,1,t] + A[3,2,t] + A[3,3,t] = Z[2,0,t]$$

ΣW2=Z2

$$W[1,0,t] + W[1,1,t] + W[1,2,t] + W[1,3,t] = Z[1,0,t]$$

$$W[1,0,t] + W[1,1,t] + W[1,2,t] + W[1,3,t] = Z[2,0,t]$$

W2[1][i][t] に関する論理積制約

$$0 \leq Z[1,i,t] + U[1,t] - 2 \cdot W[1,i,t] \leq 1$$

$$0 \leq Z[2,i,t] + U[1,t] - 2 \cdot W[1,i,t] \leq 1$$

X2 の範囲制約

$$L[1] \cdot W[1,i,t] \leq x[1,i,t] \leq L[1] \cdot U[1,t]$$

$$L[2] \cdot W[2,i,t] \leq x[2,i,t] \leq L[2] \cdot W[1,i,t]$$

$$0 \leq x[3,i,t] \leq L[3] \cdot W[2,i,t]$$

x2[0][i][t] = U2[i][t]

$$x[0,0,t] = 0$$

$$(x[0,i,t] = U[i,t])$$

x2&y2[1~3][0]==0

$$x[0,i,t] = 0$$

$$y[i,0,t] = 0$$

$$y[i,0,t] = 0$$

$$y[0,i,t] = 0$$

y2 に関する制約（U2[i, t]を使うように変更）

$$L[1] \cdot (U[1,t] + U[2,t]) \leq y[i,1,t] \leq L[1] \cdot (U[1,t] + U[2,t] + U[3,t])$$

$$L[2] \cdot U[3,t] \leq y[i,2,t] \leq L[2] \cdot (U[2,t] + U[3,t])$$

$$0 \leq y[i,2,t] \leq L[3] \cdot U[3,t]$$

$$y[0,0,t] = 0$$

$$y[i,0,t] = U[0,t]$$

z に関する制約

$$Z'[1,0,t] = 1 - Z[1,0,t]$$

$$0 \le Z'[1,0,t] + U[1,t] - 2 \cdot A[1,1,t] \le 1$$

$$0 \le Z'[1,0,t] + U[2,t] - 2 \cdot A[1,2,t] \le 1$$

$$0 \le Z'[1,0,t] + U[3,t] - 2 \cdot A[1,3,t] \le 1$$

$$0 \le Z'[2,0,t] + U[1,t] - 2 \cdot A[2,1,t] \le 1$$

$$0 \le Z'[2,0,t] + U[2,t] - 2 \cdot A[2,2,t] \le 1$$

$$0 \le Z'[2,0,t] + U[3,t] - 2 \cdot A[2,3,t] \le 1$$

(i=1,2,3)↓

$$L[1] \cdot (A[i,2,t] + A[i,3,t]) \le y[i,1,t]$$

$$L[1] \cdot (A[i,1,t] + A[i,2,t] + A[i,3,t]) \ge y[i,1,t]$$

$$L[3] \cdot A[i,3,t] \le y[i,2,t]$$

$$L[2] \cdot (A[i,3,t] + A[i,2,t]) \le y[i,2,t]$$

$$0 \le y[i,3,t]$$

$$L[3] \cdot A[i,3,t] \ge y[i,3,t]$$

$$0 \le Z[1,0,t] + (1 - Z[2,0,t]) - 2 \cdot Z'[2,0,t] \le 1$$

$$0 \le Z[i-1,0,t] + U[1,t] - 2 \cdot A[i,1,t] \le 1$$

$$0 \leq Z[i-1,0,t] + U[2,t] - 2 \cdot A[i,2,t] \leq 1$$

$$0 \leq Z[i-1,0,t] + U[3,t] - 2 \cdot A[i,3,t] \leq 1$$